



Inteligencia Artificial y Aprendizaje Automático

Sesión 4: Redes Neuronales
Convolucionales

Presentan:

Dr. Ulises Olivares Pinto

Fernando Ramírez González

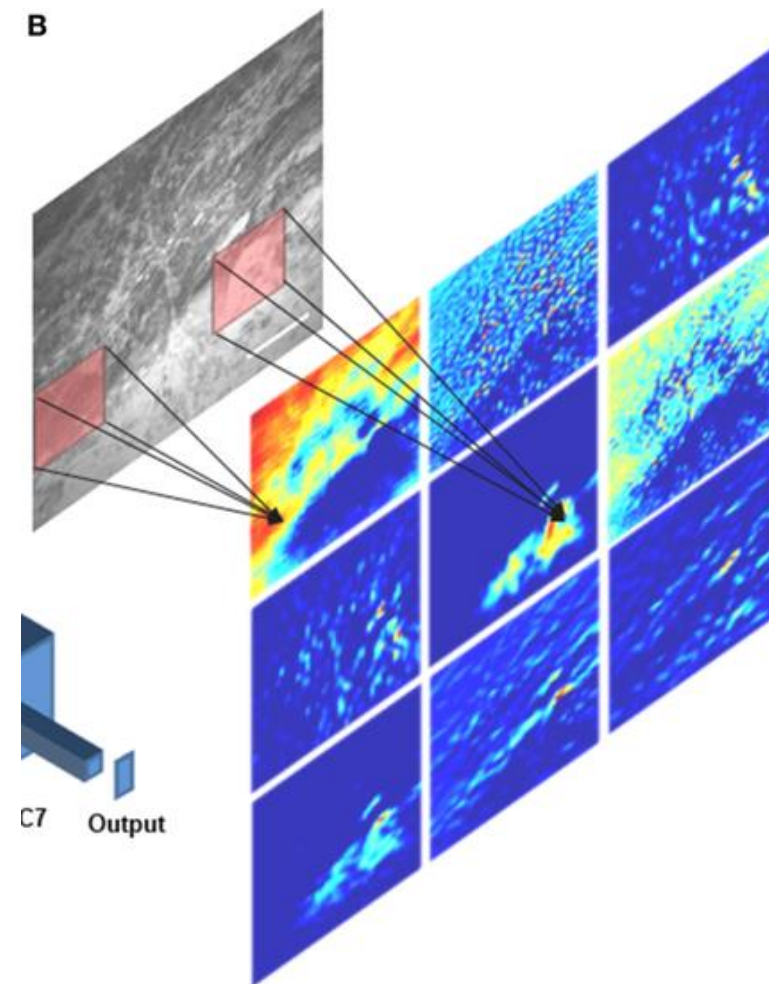


HPC
DATA LAB

Contenido

Redes Neuronales convolucionales

- Definiciones
- Convolución
- Diseño de arquitecturas
- Ejercicios



Las imágenes son números



157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
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What the computer sees

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183	202	237	145	0	0	12	108	200	138	243	236
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Image



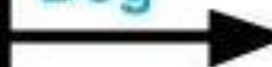
Neural Network

Cat



0.97

Dog



0.01

Other



0.02







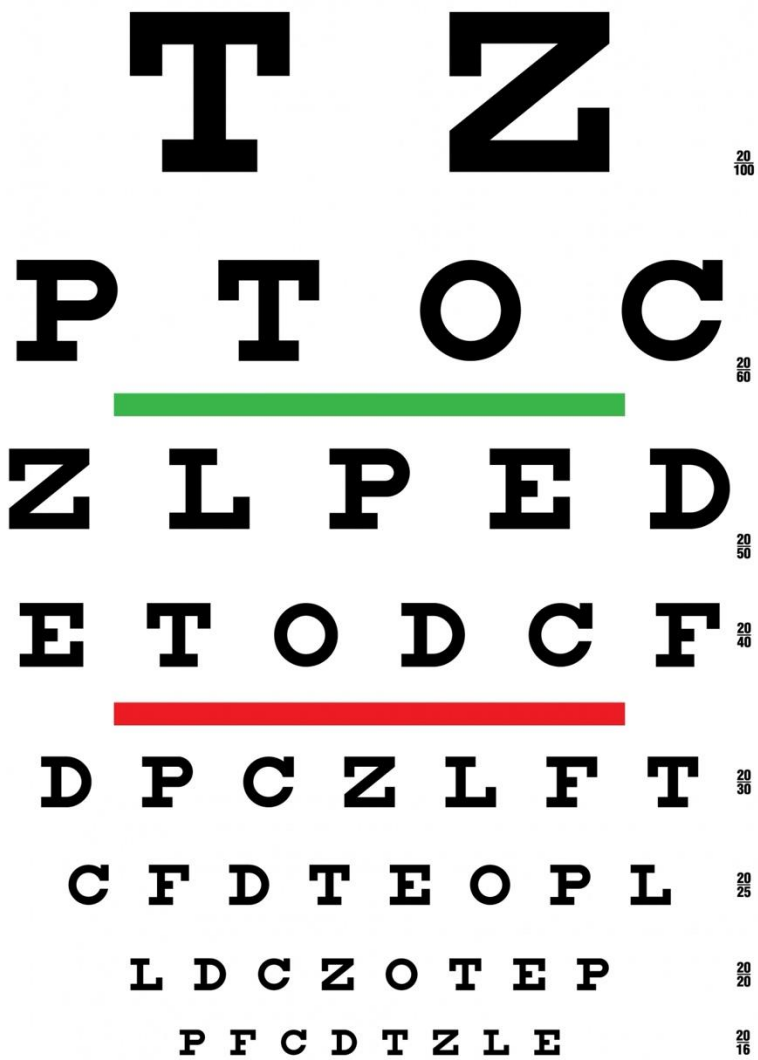




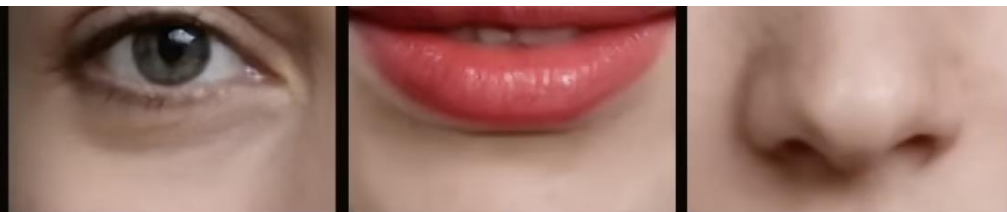
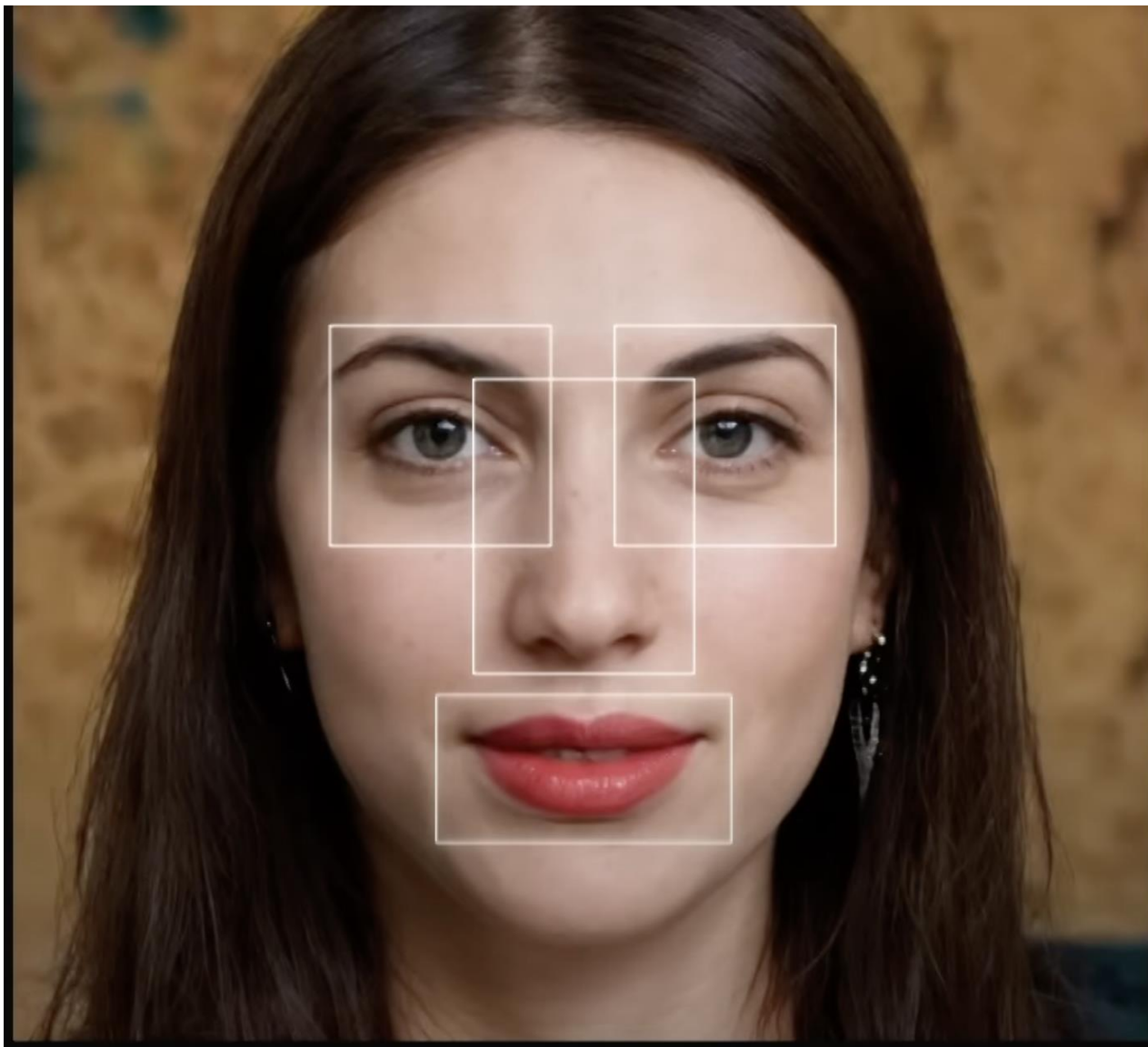






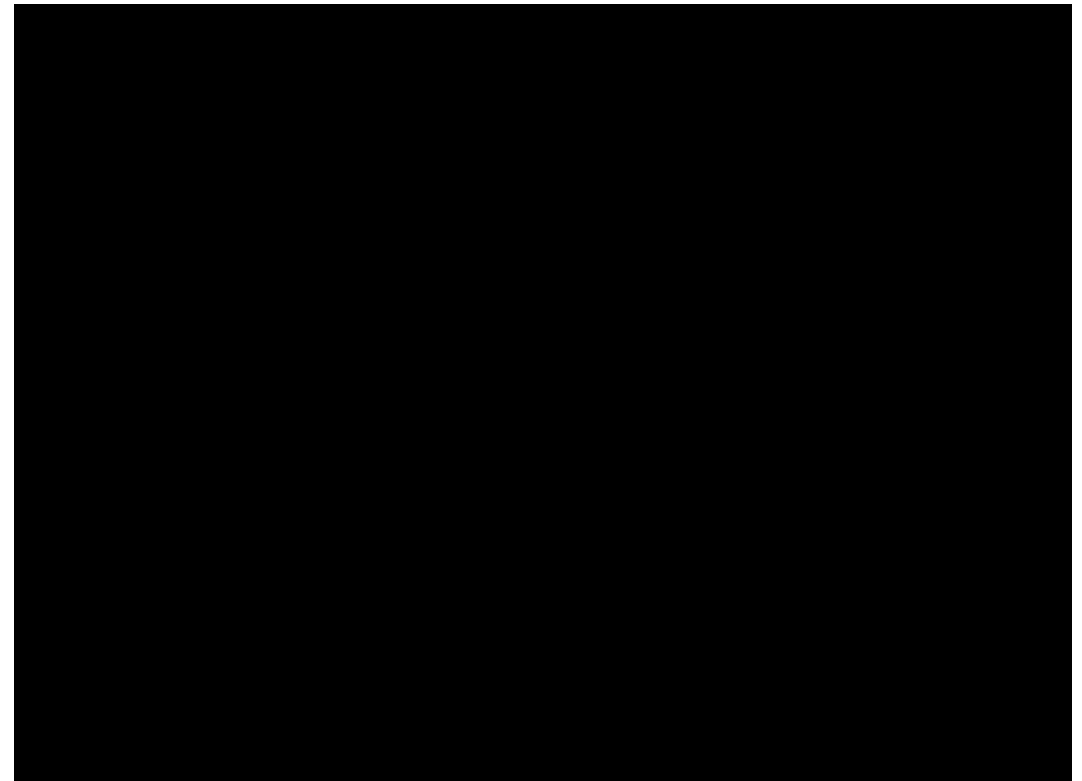






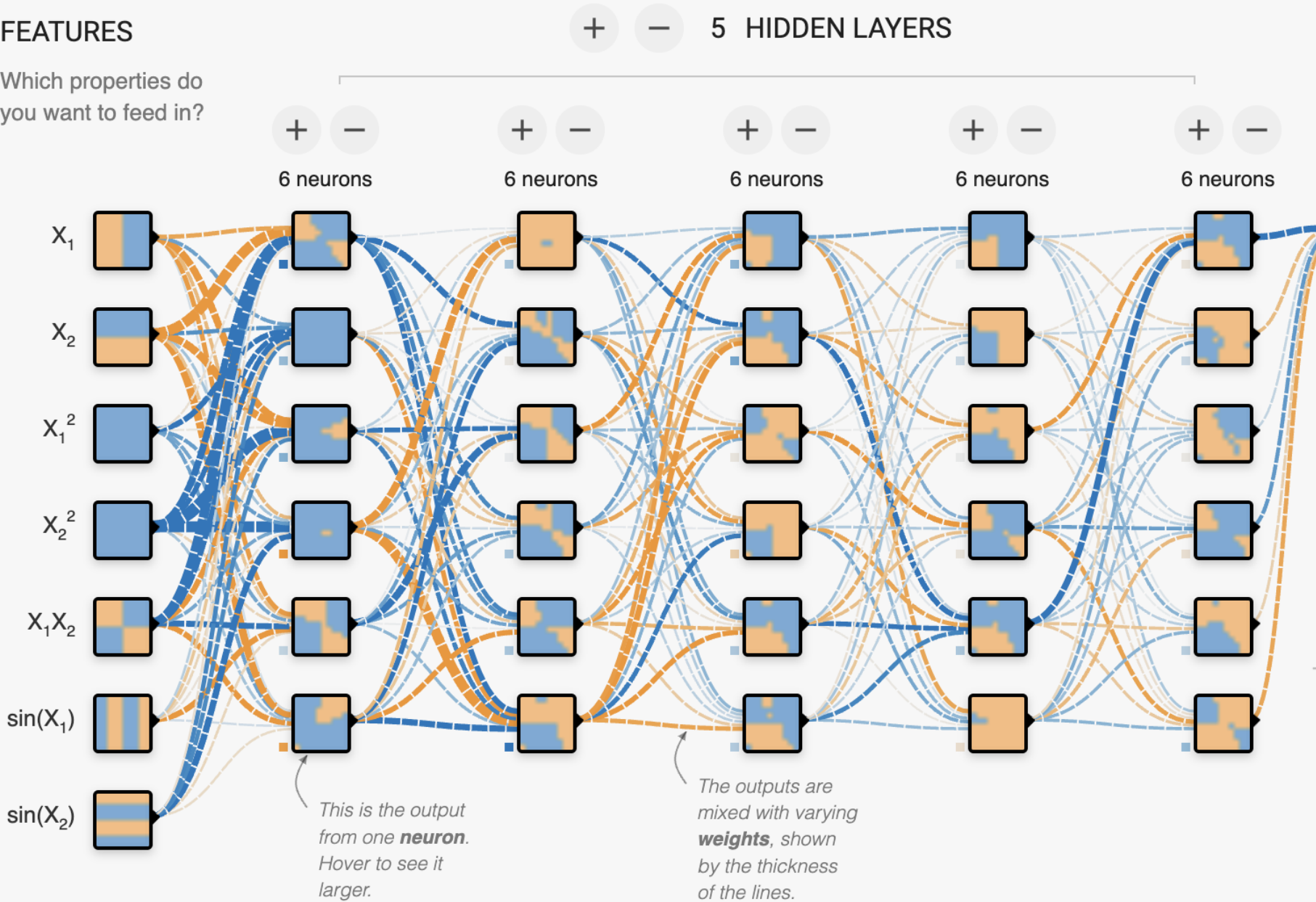


Redes Neuronales Convolucionales (Yann LeCun, 1989)



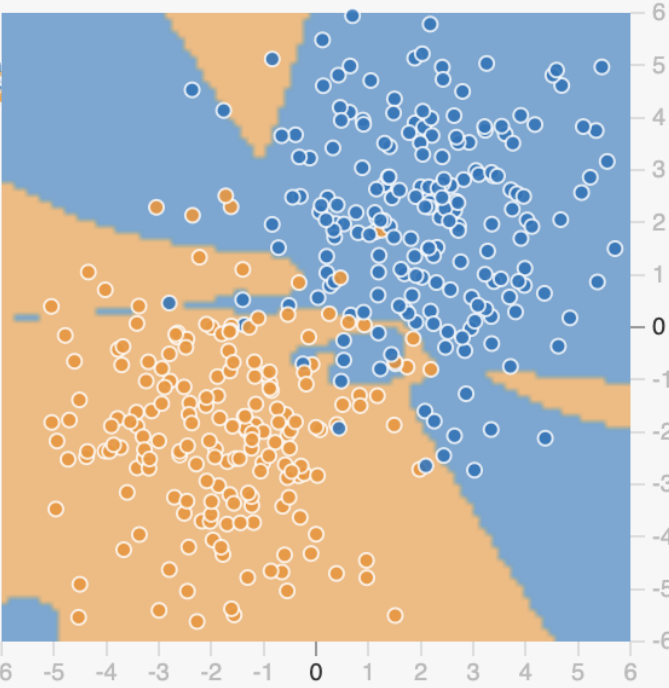
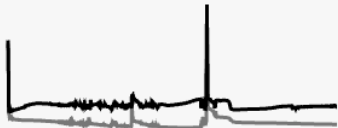
FEATURES

Which properties do you want to feed in?



OUTPUT

Test loss 0.150
Training loss 0.048



Colors shows data, neuron and weight values.

-1 0 1

☐ Show test data ☒ Discretize output



28x28 pixels

`.flatten(img)` →



784x1 pixels

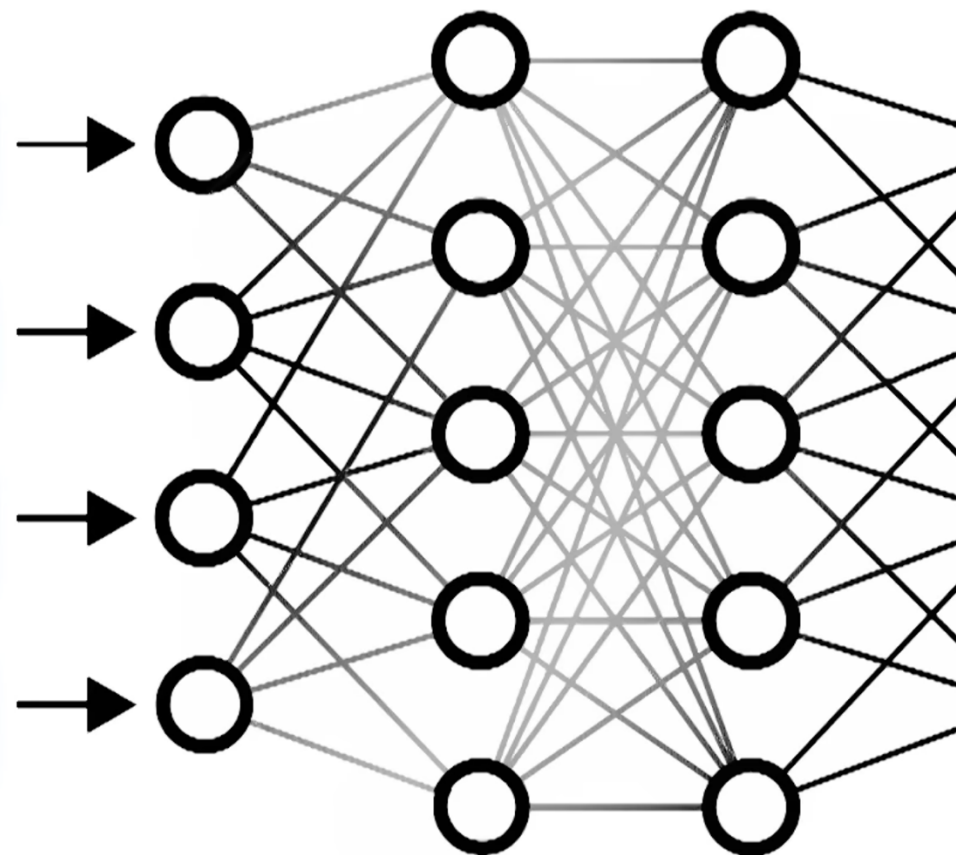
pixel 0

pixel 1

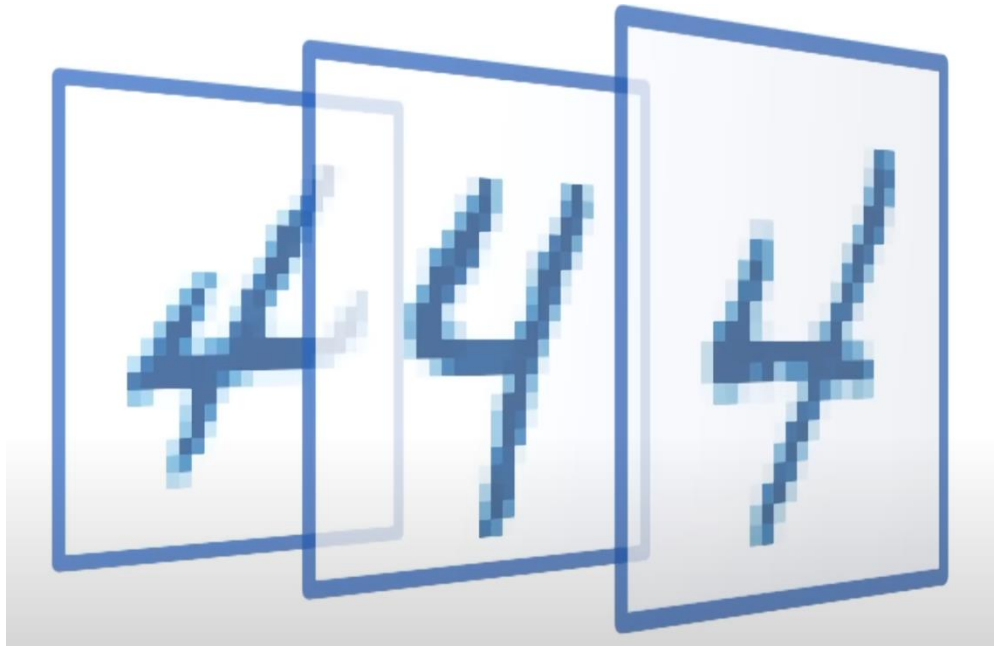
pixel 2

pixel 3

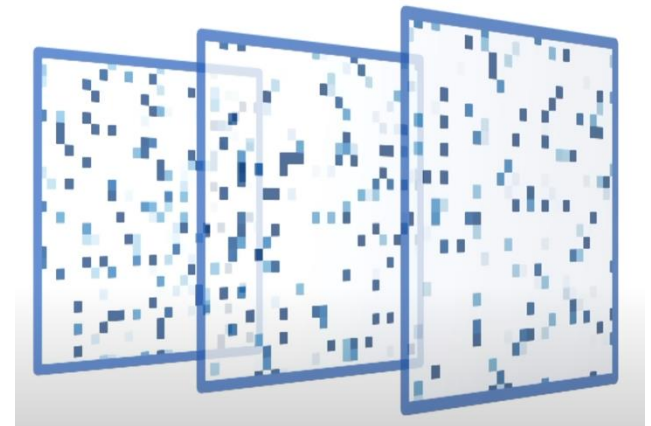
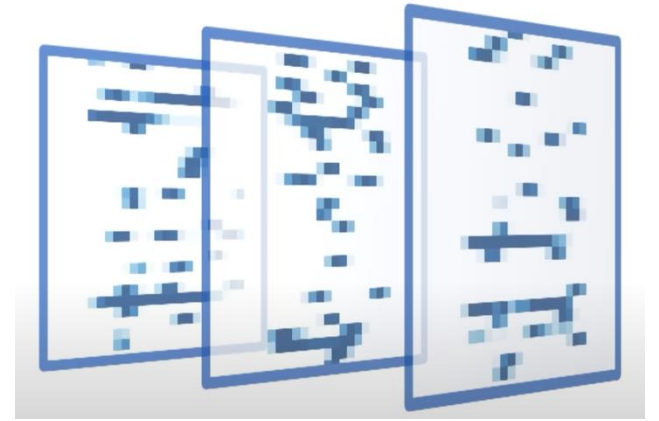
...



Limitantes de las Redes Neuronales



=



Filtros de convolución





Original



X-Pro II



Lomo-fi



Earlybird



Sutro



Lily



Toaster



Brannan



Inkwell



Walden



Hefe



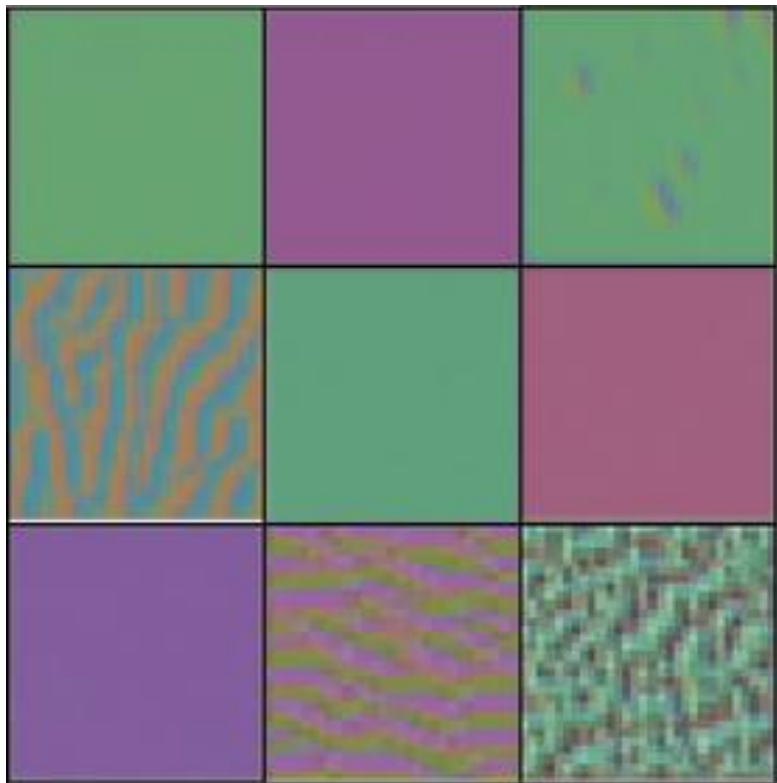
Apollo



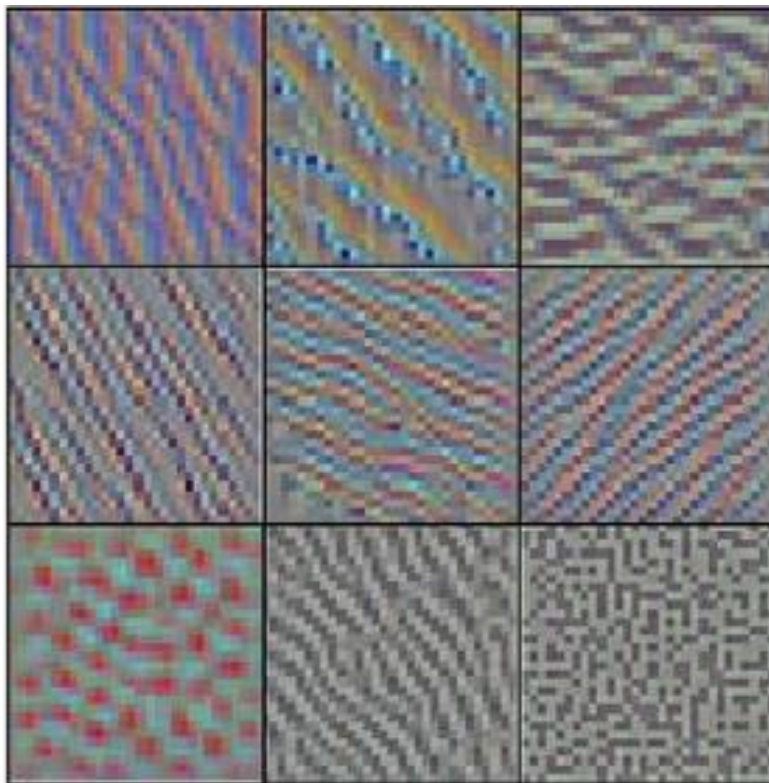
Filtros de convolución

<i>Original</i>	<i>Gaussian Blur</i>	<i>Sharpen</i>	<i>Edge Detection</i>
$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$
			

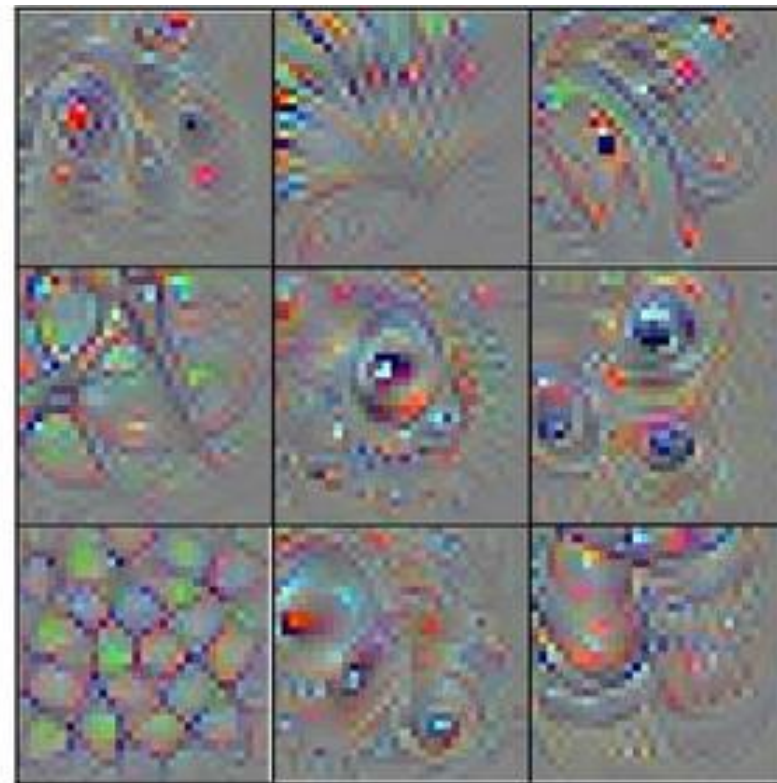
Extracción de características



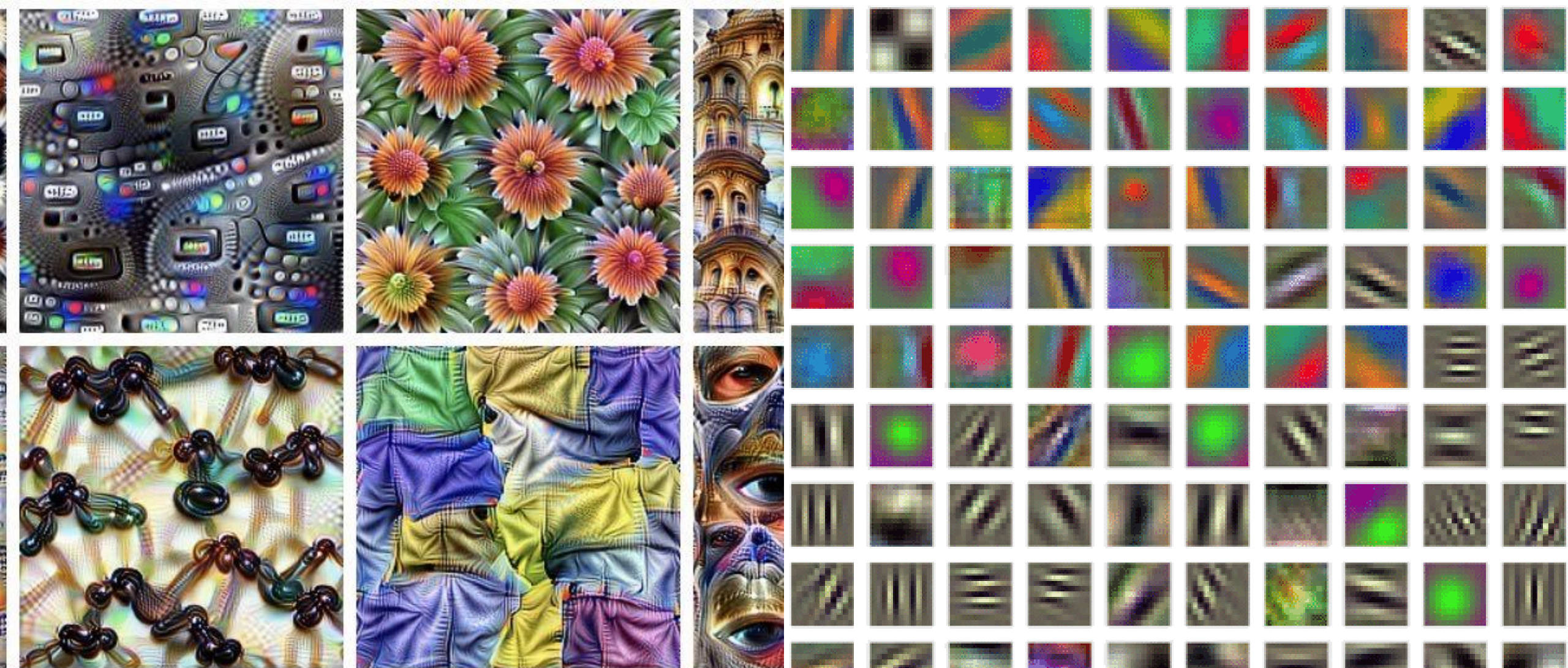
(a) Color



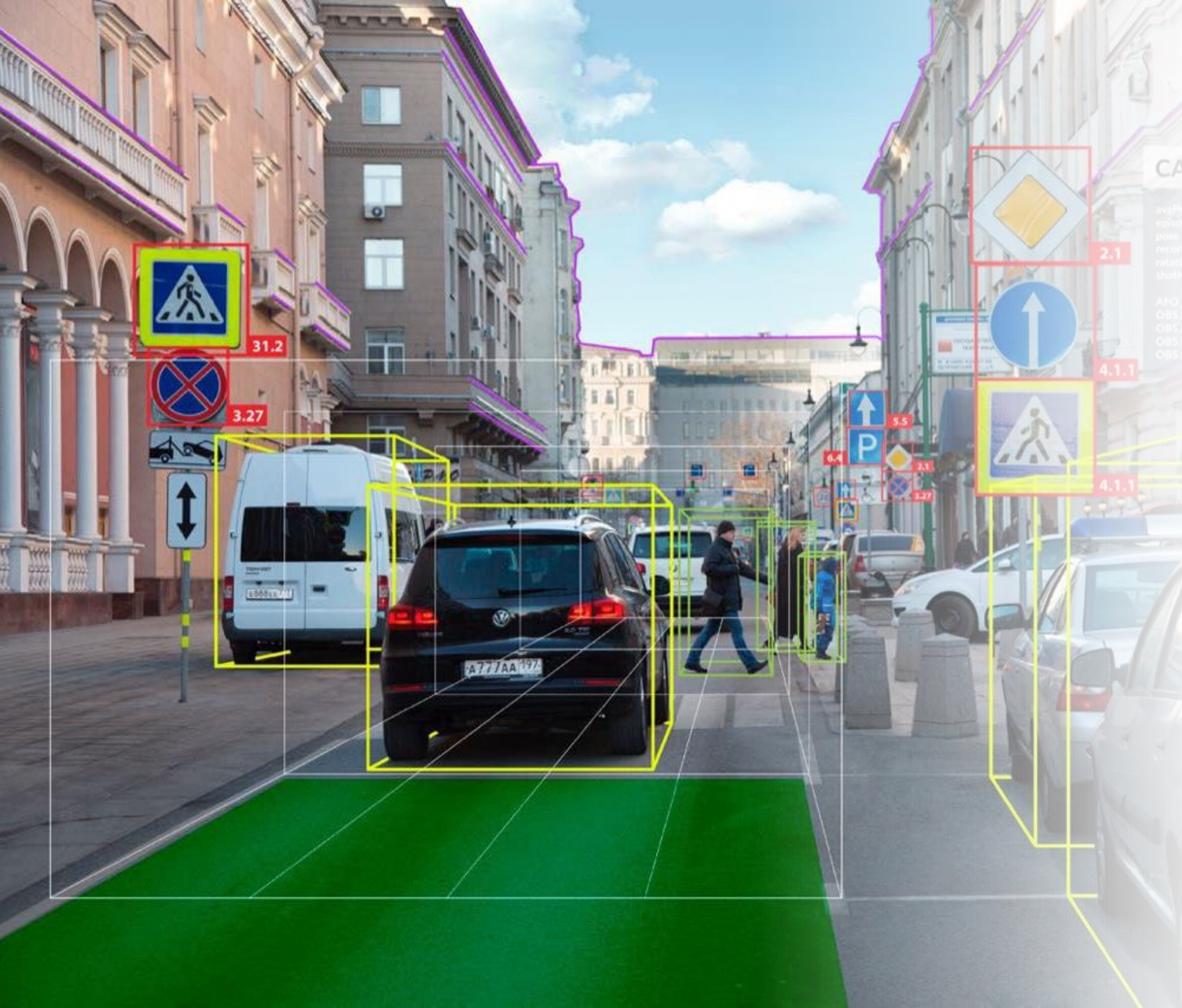
(b) Texture



(c) Structure



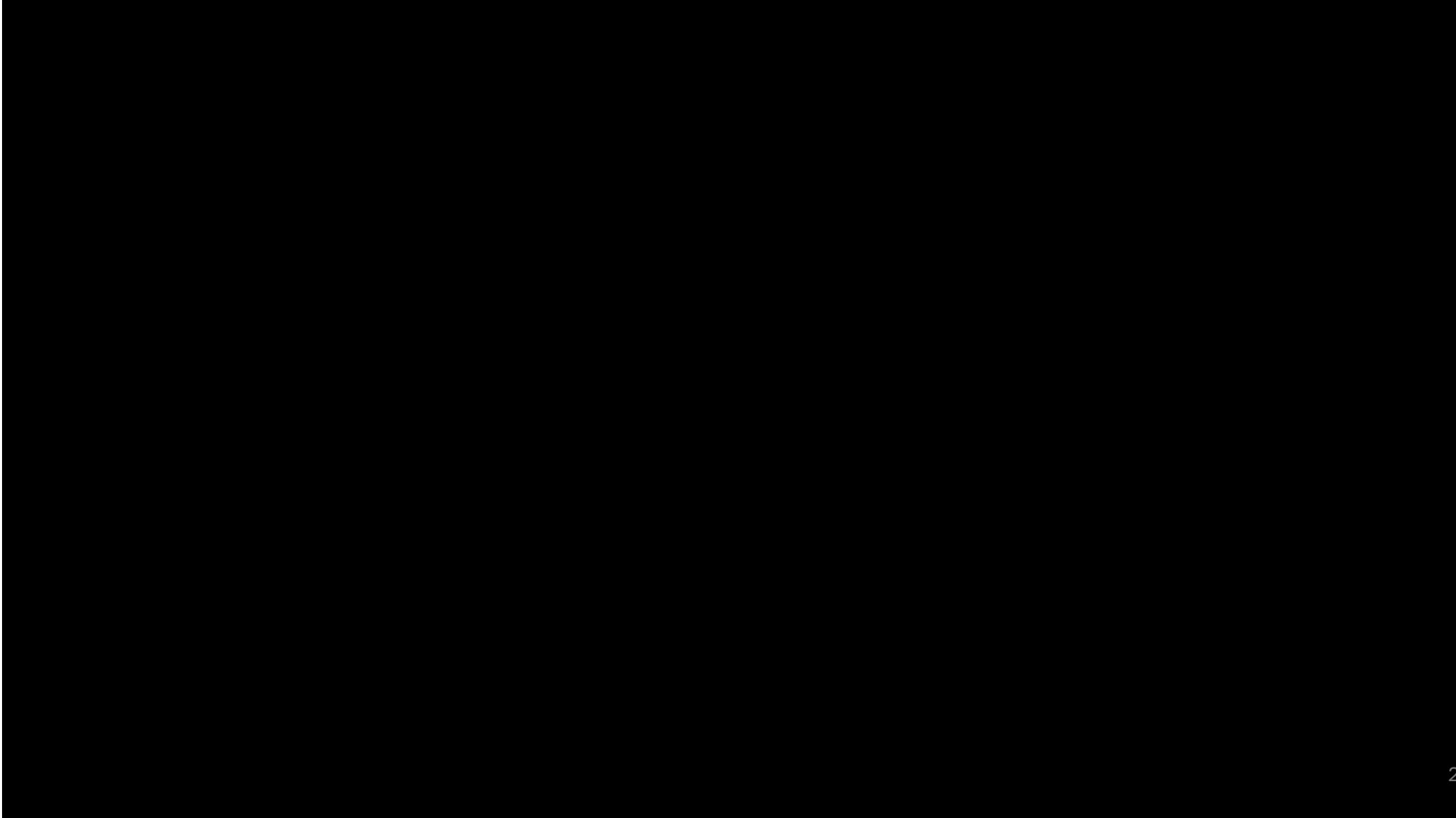
Filtros de convolución



¿Dónde están los
objetos, qué son los
objetos, qué acciones
están ocurriendo?

Predecir y anticipar
eventos en el mundo

Teacheble Machine



Teachable Machine

<https://teachablemachine.withgoogle.com/>

New Project

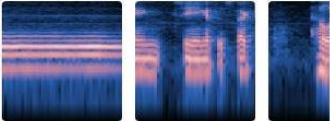
Open an existing project from Drive.

Open an existing project from a file.



Image Project

Teach based on images, from files or your webcam.



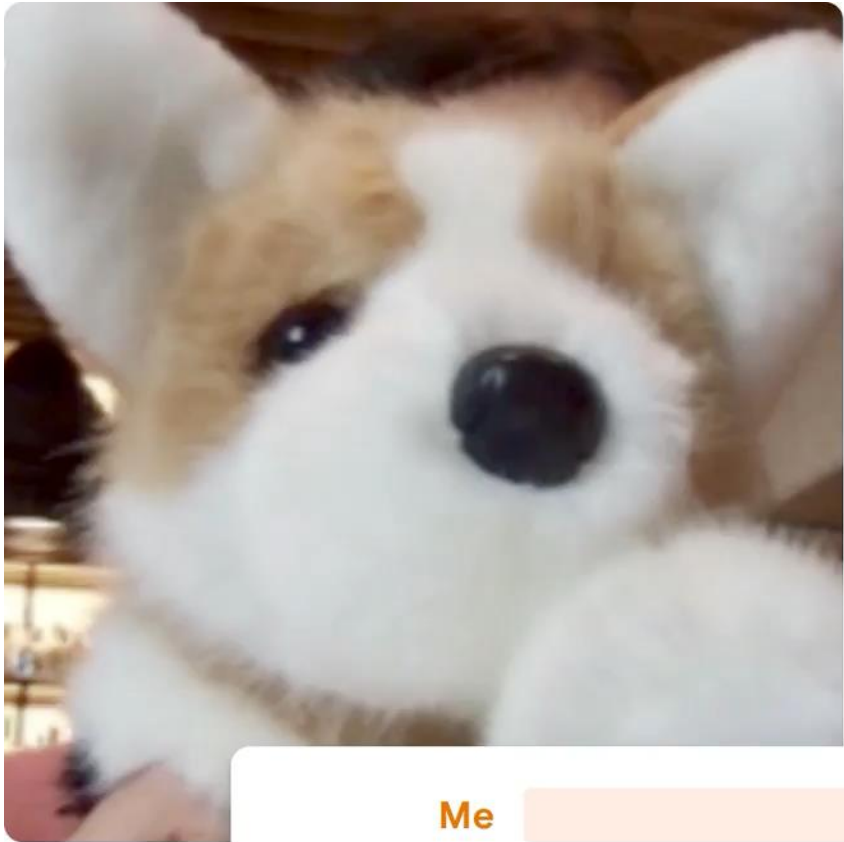
Audio Project

Teach based on one-second-long sounds, from files or your microphone.



Pose Project

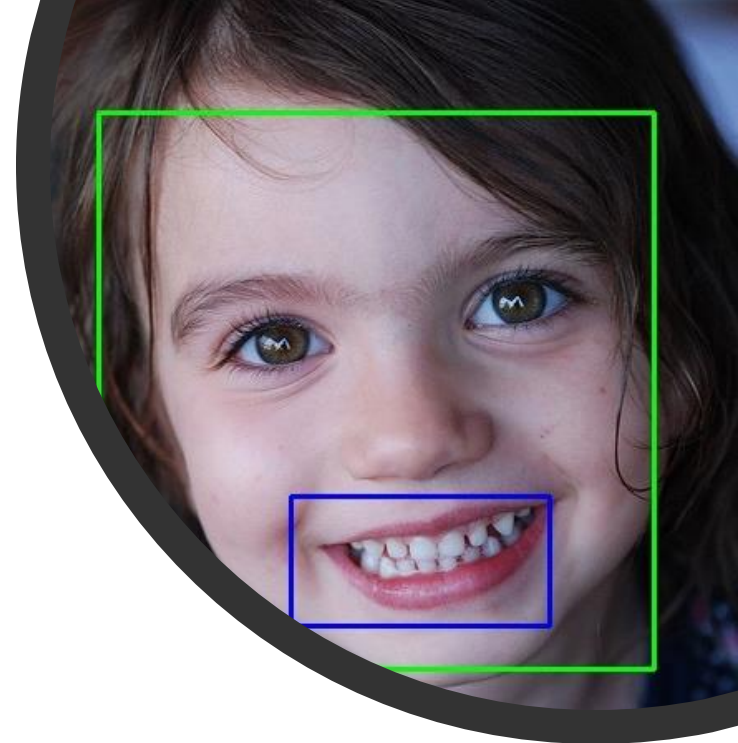
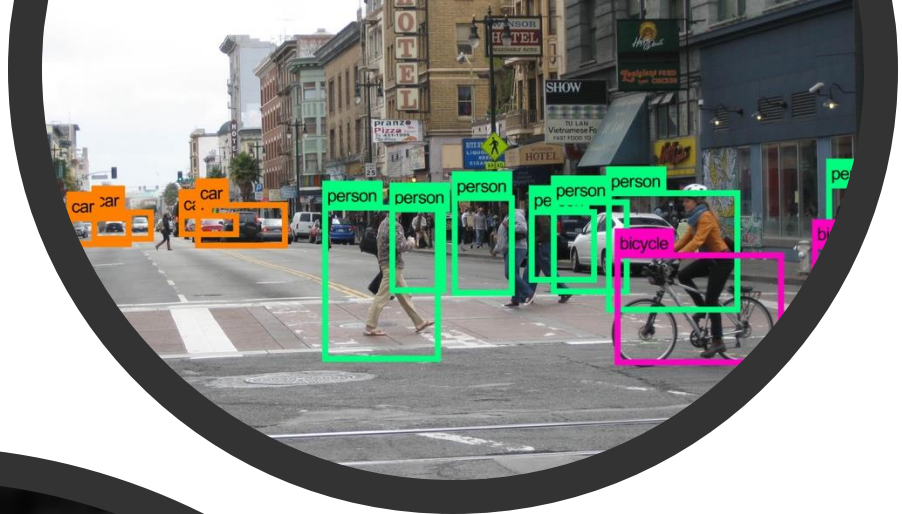
Teach based on images, from files or your webcam.



Me

Me + Dog <3

20%



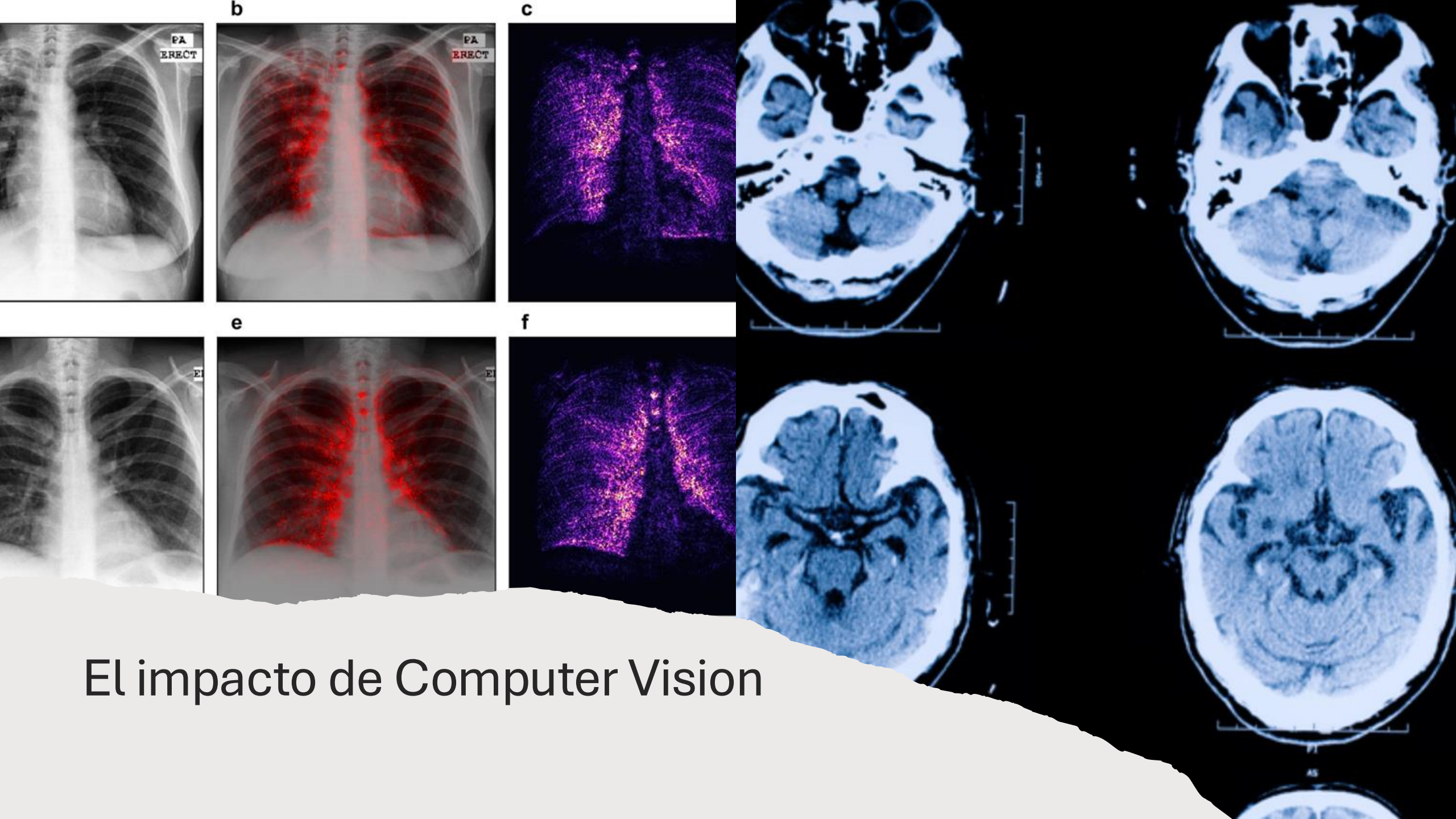
El impacto de Computer Vision





OpenAI Microscope

Open AI Microscope
<https://microscope.openai.com>



El impacto de Computer Vision

RESOURCE | VOLUME 172, ISSUE 5, P1122-1131.E9, FEBRUARY 22, 2018



PDF [4 MB]



Figures



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Reprints



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Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning

[Daniel S. Kermany](#) ¹⁴ • [Michael Goldbaum](#) ¹⁴ • [Wenjia Cai](#) ¹⁴ • ... [M. Anthony Lewis](#) • [Huimin Xia](#) •

[Kang Zhang](#) ¹⁵ • [Show all authors](#) • [Show footnotes](#)

[Open Archive](#) • DOI: <https://doi.org/10.1016/j.cell.2018.02.010> •



Check for updates

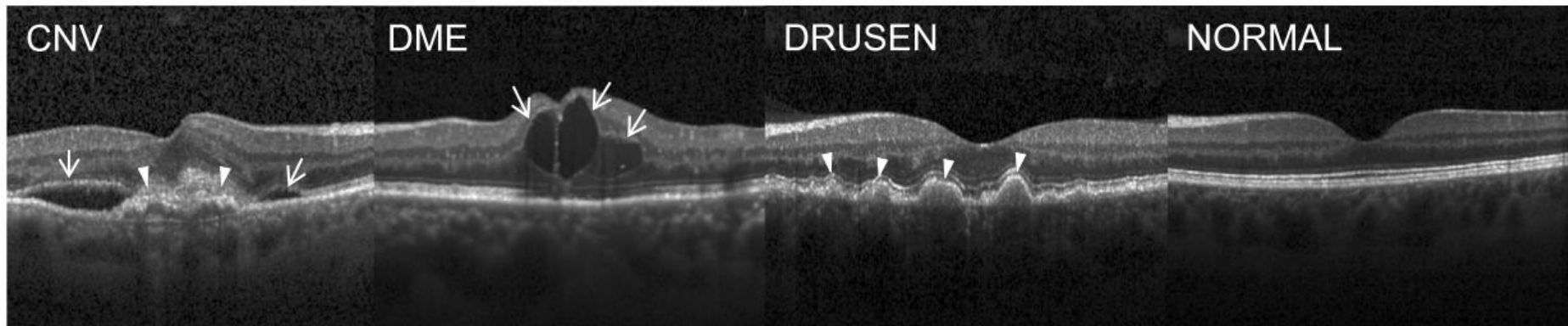
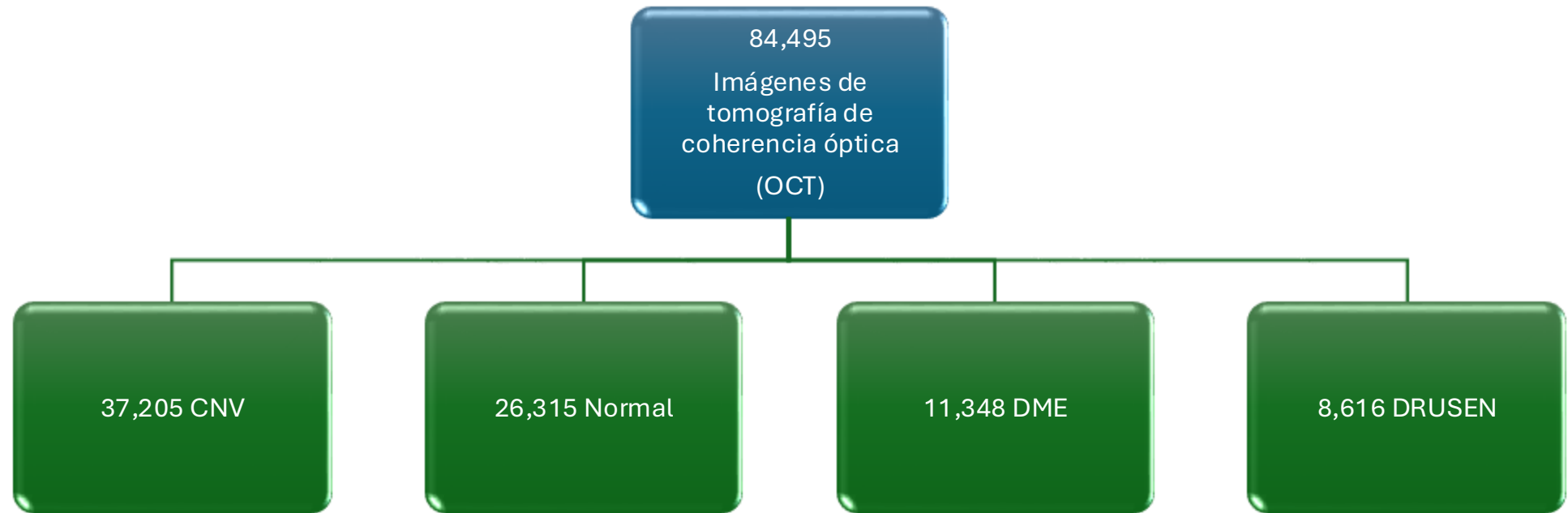


PlumX Metrics

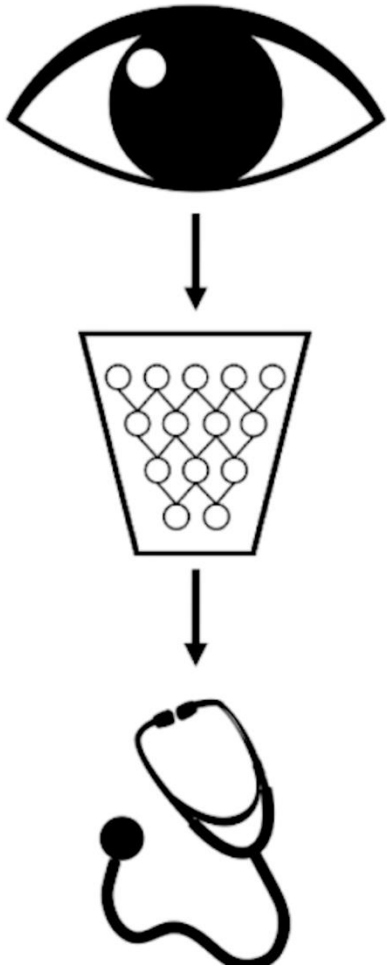
Ejemplo

Ejemplo:

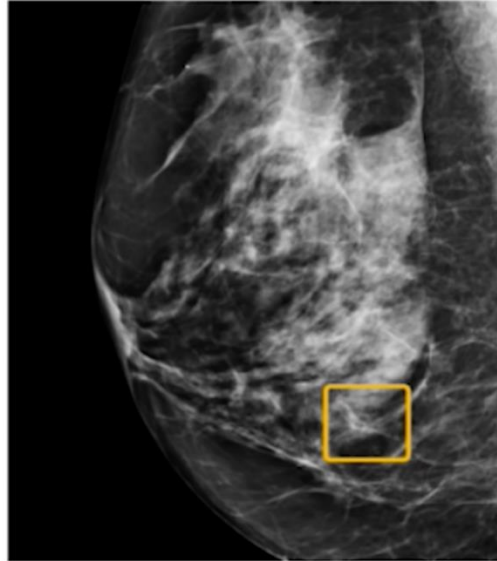
Datos para el estudio (Entrenamiento)



Aplicaciones: Salud



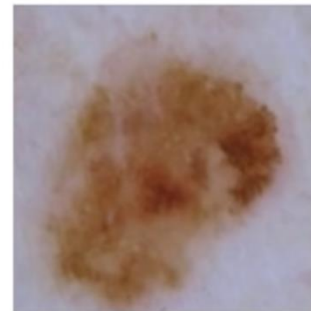
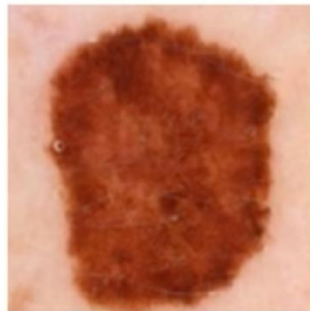
Breast cancer



COVID-19



Skin cancer





Lo que puede ver una computadora



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Las imágenes son números



Input Image



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195	206	123	207	177	121	123	200	175	13	96	218

Pixel Representation



classification

Lincoln

Washington

Jefferson

Obama

$$\begin{bmatrix} 0.8 \\ 0.1 \\ 0.05 \\ 0.05 \end{bmatrix}$$

Tareas en visión computacional

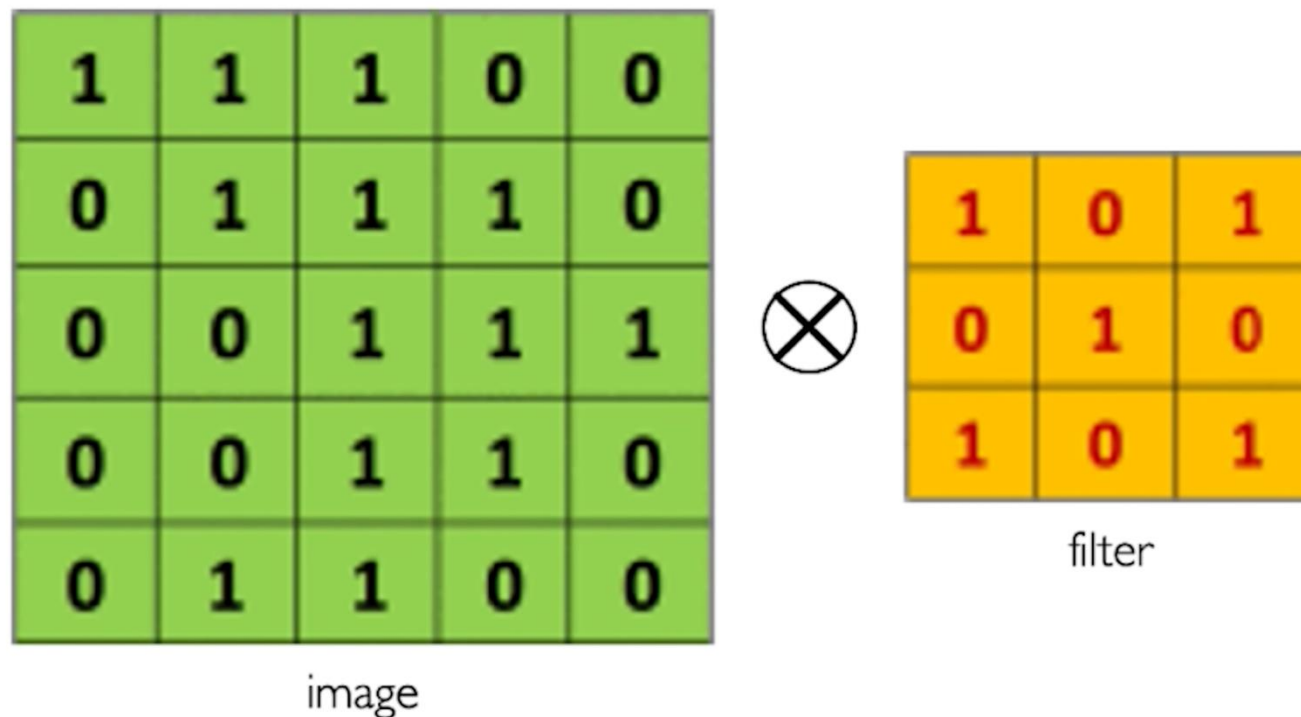
Fuente: Introduction to Deep Learning MIT OpenCourseWare, 2021

Operación de convolución



Operación de convolución

Suppose we want to compute the convolution of a 5x5 image and a 3x3 filter:



We slide the 3x3 filter over the input image, element-wise multiply, and add the outputs...

Operación de convolución

1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0



1	0	1
0	1	0
1	0	1

filter

=

4		

feature map

Operación de convolución

1	1 _{x1}	1 _{x0}	0 _{x1}	0
0	1 _{x0}	1 _{x1}	1 _{x0}	0
0	0 _{x1}	1 _{x0}	1 _{x1}	1
0	0	1	1	0
0	1	1	0	0



1	0	1
0	1	0
1	0	1

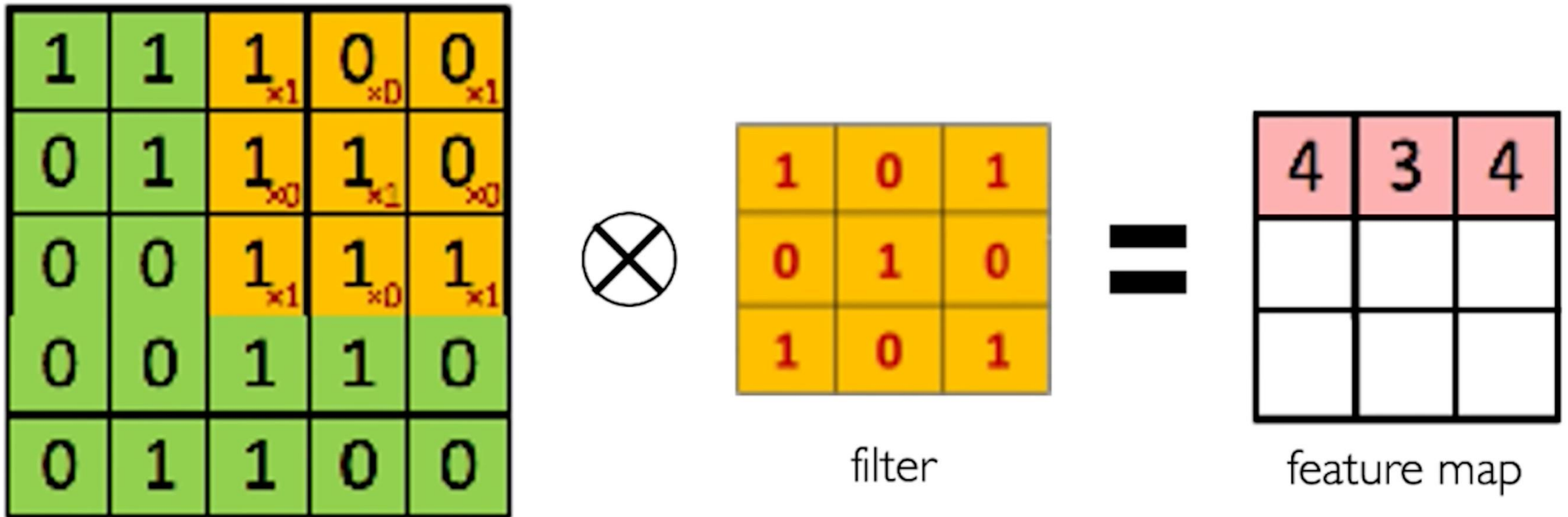
filter



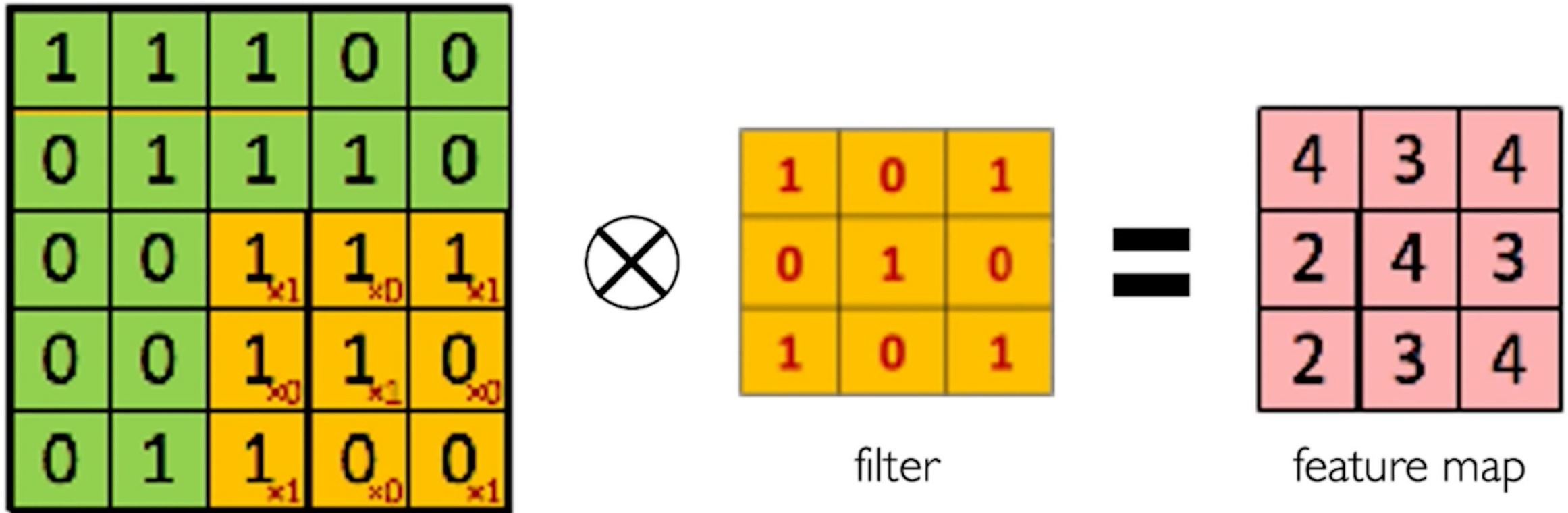
4	3	

feature map

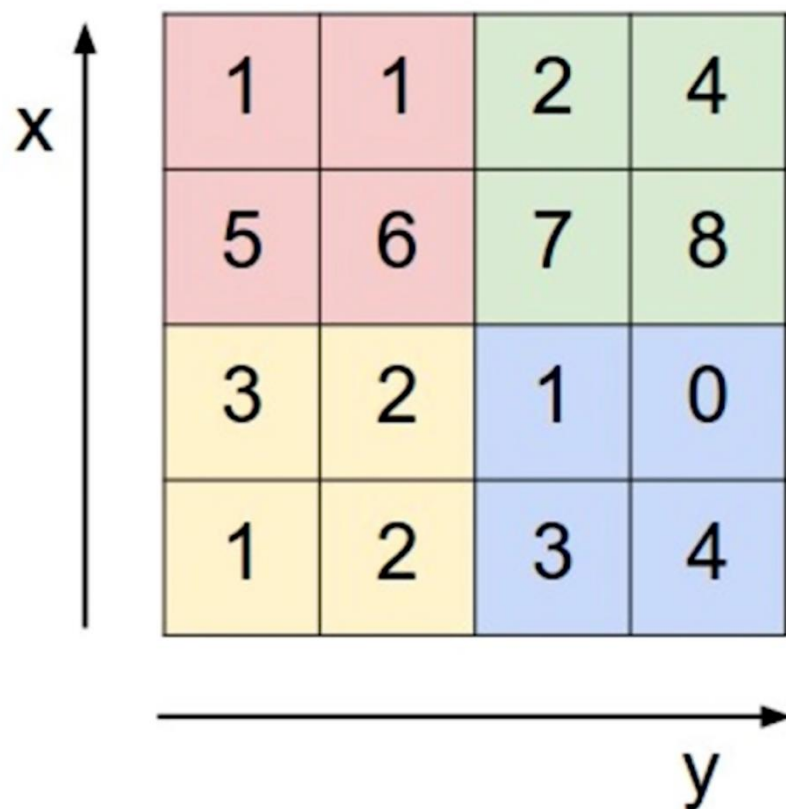
Operación de convolución



Operación de convolución



Pooling (agrupación)



1	1	2	4
5	6	7	8
3	2	1	0
1	2	3	4

max pool with 2x2 filters
and stride 2

```
tf.keras.layers.MaxPool2D(  
    pool_size=(2,2),  
    strides=2  
)
```



6	8
3	4

- 1) Reduced dimensionality
- 2) Spatial invariance

How else can we downsample and preserve spatial invariance?

Convoluciones - Actividad en equipo

- Actividad:
 - Experimentar aplicando Kernels de convolución a imágenes.
 - <https://setosa.io/ev/image-kernels/>

Image Kernels

Explained Visually

 Post
 Me gusta 440
 Com

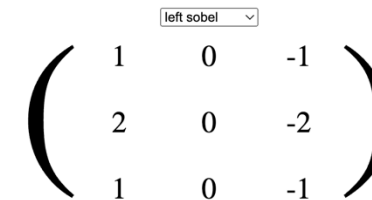
By Victor Powell

An image kernel is a small matrix used to apply effects like the ones you might find in Photoshop or Gimp, such as blurring, sharpening, outlining or embossing. They're also used in machine learning for 'feature extraction', a technique for determining the most important portions of an image. In this context the process is referred to more generally as "convolution" (see: [convolutional neural networks](#).)

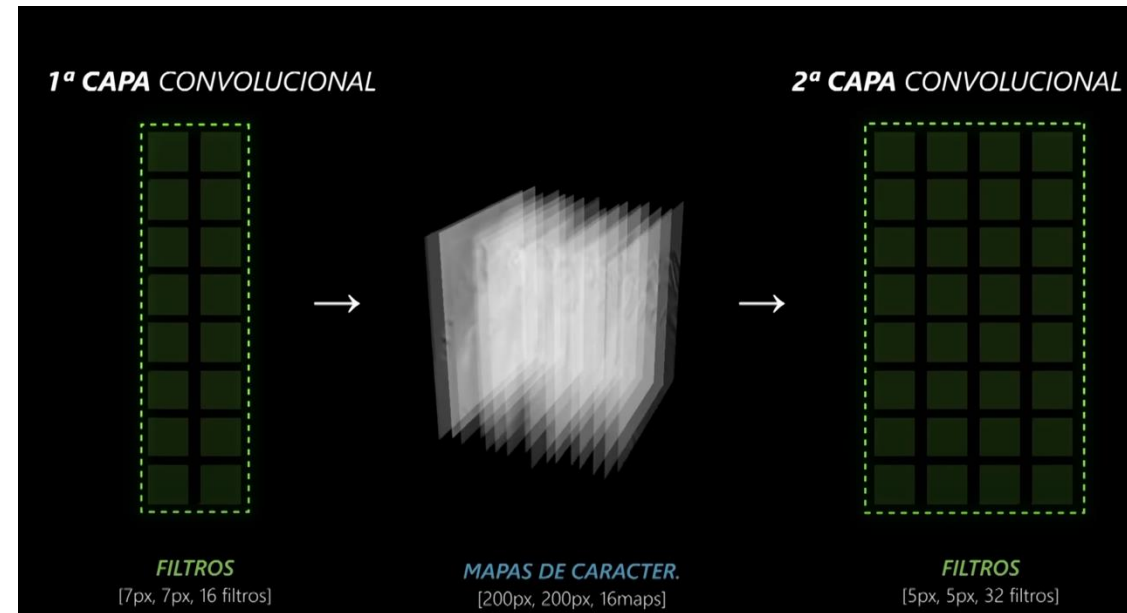
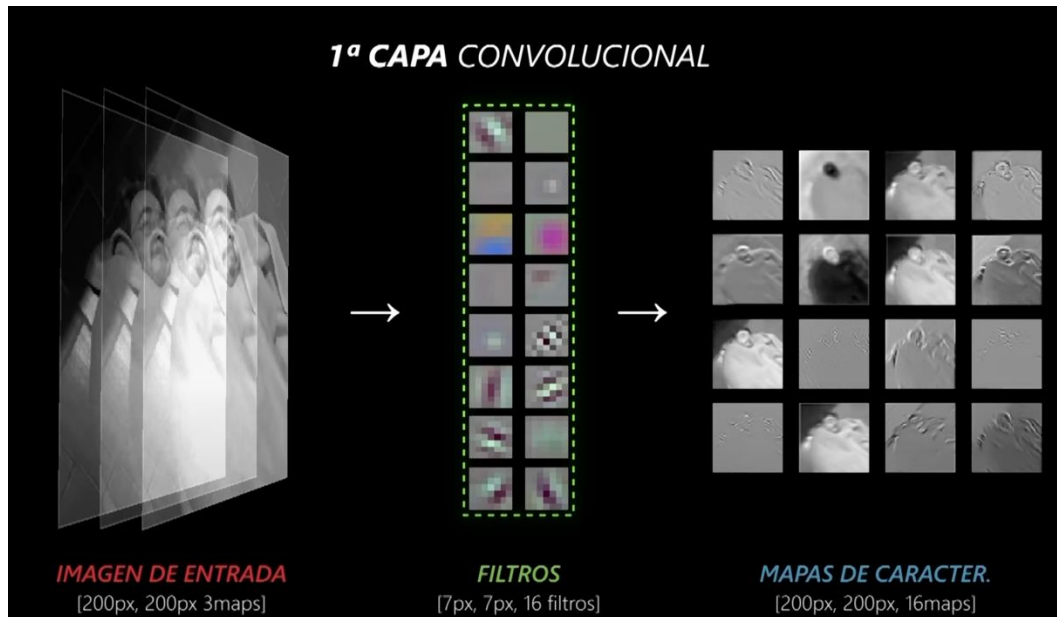
To see how they work, let's start by inspecting a black and white image. The matrix on the left contains numbers, between 0 and 255, which each correspond to the brightness of one pixel in a picture of a face. The large, granulated picture has been blown up to make it easier to see; the last image is the "real" size.



Let's walk through applying the following 3x3 **left sobel** kernel to the image of a face from above.



Red Neuronal Convolucional

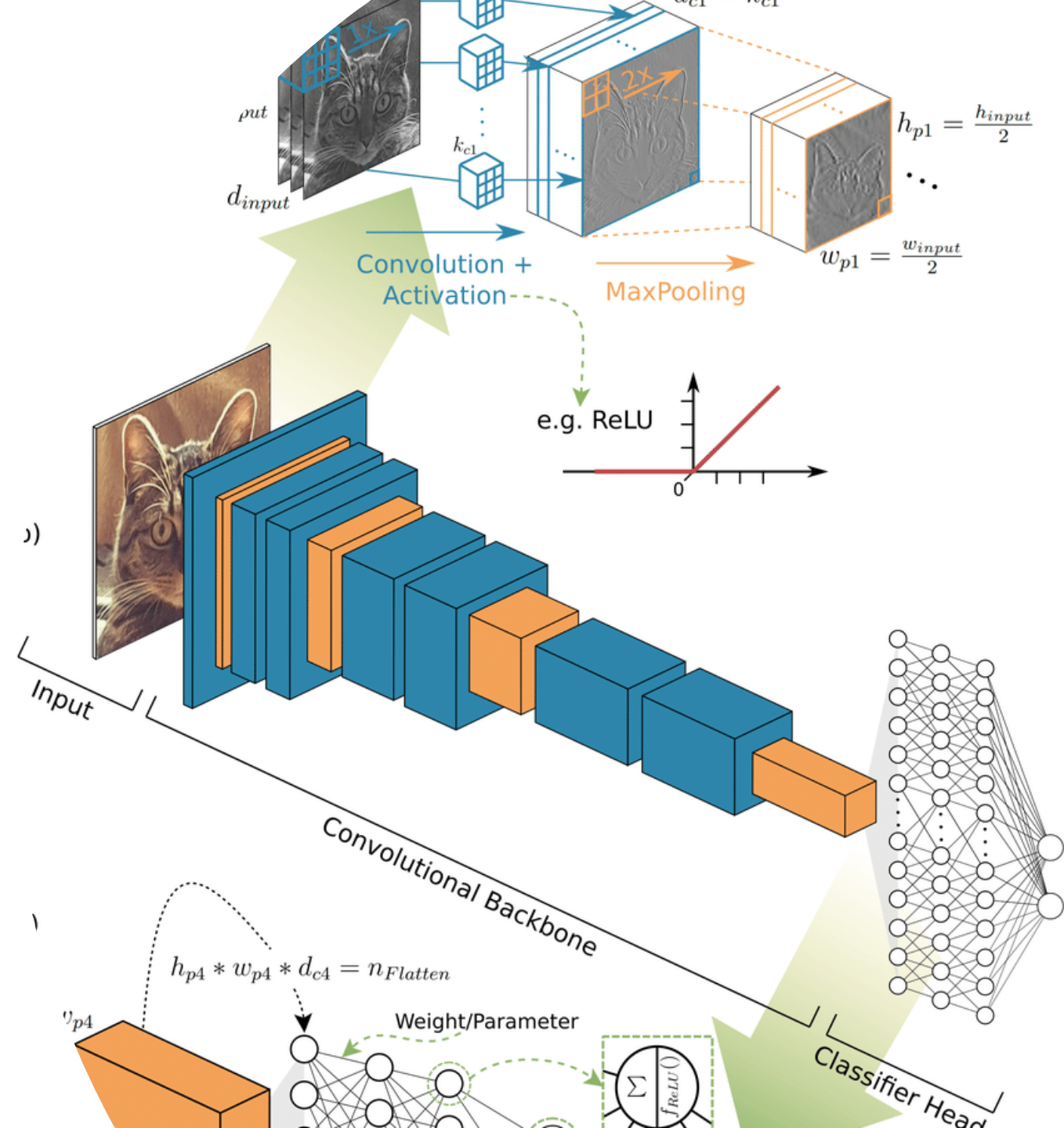


Convoluciones y Deep Learning



Redes Neuronales Convolucionales (CNNs)

Deep Learning



CNNs para clasificación

1. Convolución:

Aplicar filtros para generar mapas de características

2. Introducir no linealidad (función de activación):

Usualmente se utiliza la función RELU

3. Agrupación:

Agrupar los datos de cada mapa de características.



```
tf.keras.layers.Conv2D
```

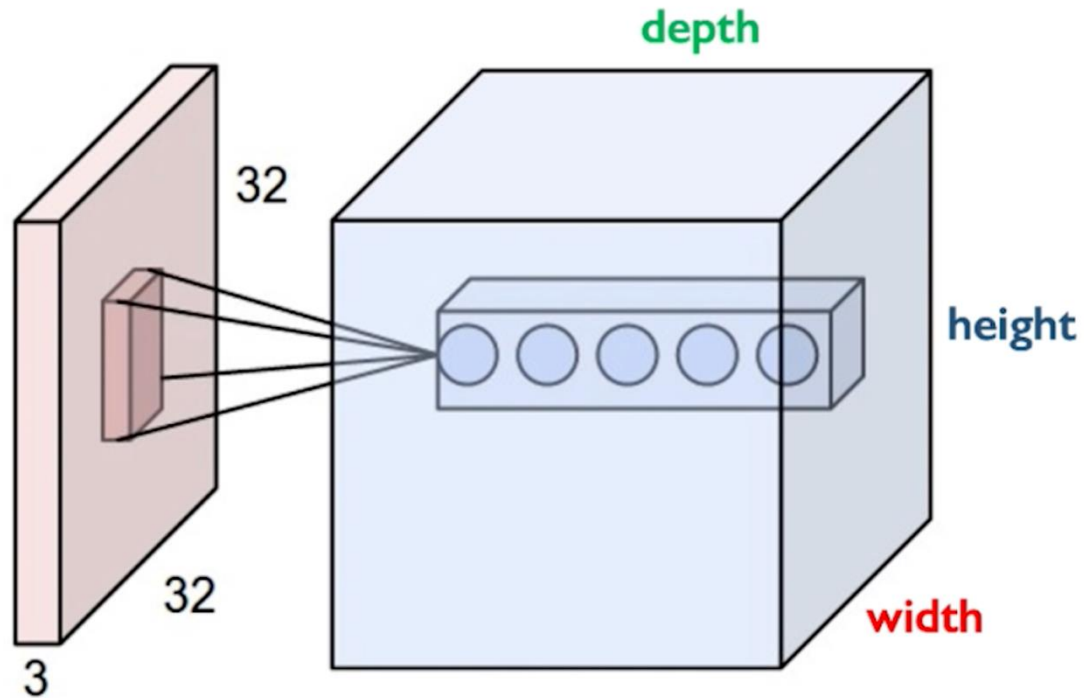


```
tf.keras.activations.*
```



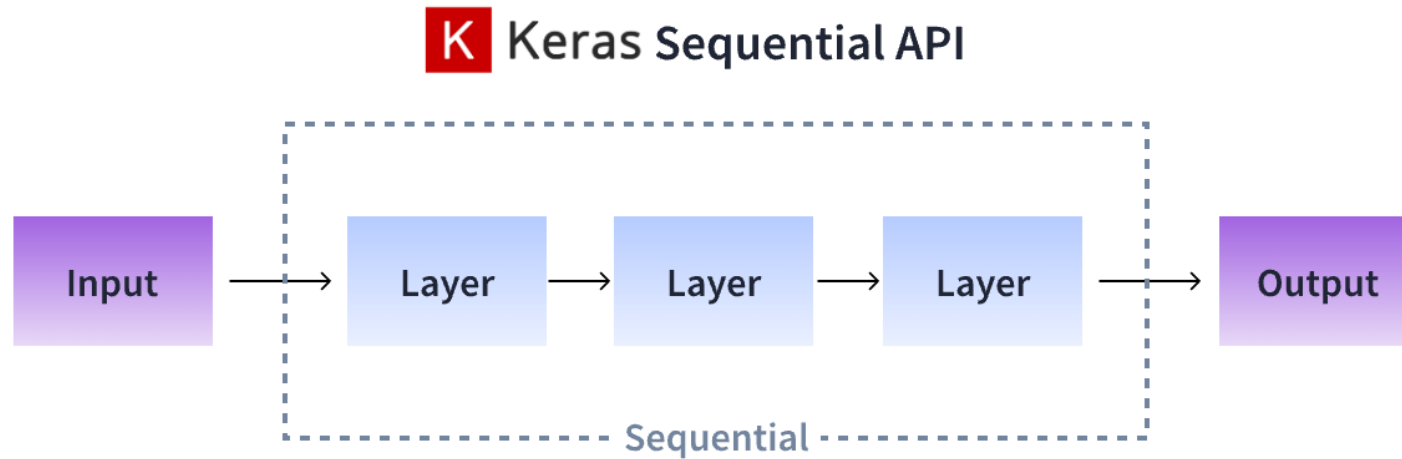
```
tf.keras.layers.MaxPool2D
```

CNNs: Arreglo espacial salidas



```
tf.keras.layers.Conv2D( filters=d, kernel_size=(h,w), strides=s )
```


Apilando capas Convolucionales



```
1 model = models.Sequential()
2 model.add(layers.Conv2D(32, (3, 3), activation='relu', input_shape=(32, 32, 3)))
3 model.add(layers.MaxPooling2D((2, 2)))
4 model.add(layers.Conv2D(64, (3, 3), activation='relu'))
5 model.add(layers.MaxPooling2D((2, 2)))
6 model.add(layers.Conv2D(64, (3, 3), activation='relu'))
7
8 # Añadiendo capas densas en el final
9 model.add(layers.Flatten())
10 model.add(layers.Dense(64, activation='relu'))
11 model.add(layers.Dense(10))
```

Demo 1: CNN y Visualización de características

CIFAR - 10

airplane



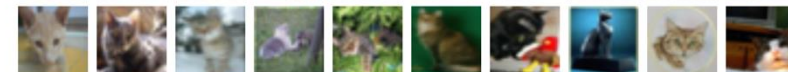
automobile



bird



cat



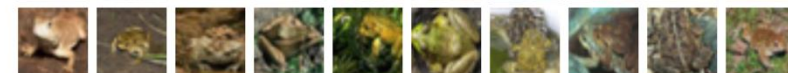
deer



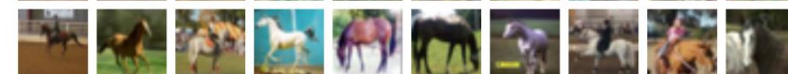
dog



frog



horse



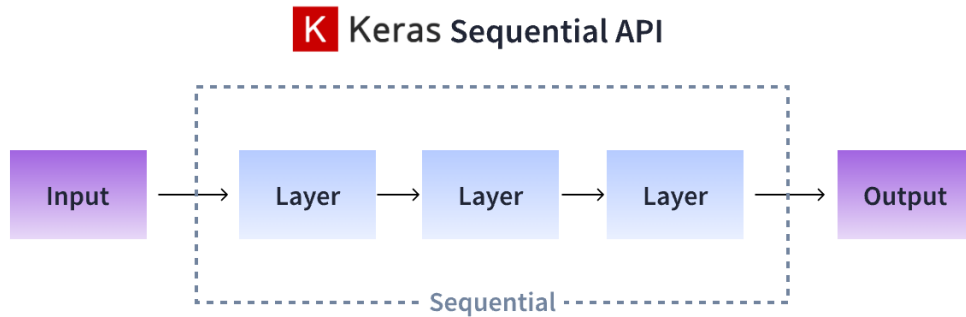
ship



truck



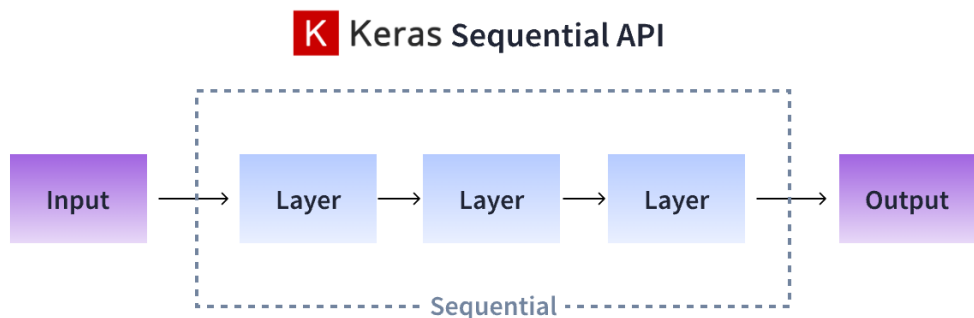
Ejercicios de programación



- **Objetivo:** Familiarizarse con la estructura básica de una red neuronal convolucional (CNN) y el proceso de entrenamiento utilizando un conjunto de datos simple como MNIST.

1. Agrega capas convolucionales y de pooling al modelo secuencial.
2. Visualiza los filtros e imágenes resultantes.

Ejercicios de programación



- **Objetivo:** Familiarizarse con la estructura básica de una red neuronal convolucional (CNN) y el proceso de entrenamiento utilizando un conjunto de datos simple como MNIST.



1. Modifica el Código para clasificar el conjunto de datos de MNIST de dígitos escritos a mano.
2. Visualizar Kernels y mapas de características.